

THE SCIENTIFIC EVIDENCE FOR VEGANISM

A deep evidence library on animals, nutrition, agriculture, ecology, public health, food systems, economics and social impacts

ABOUT THIS LIBRARY

This is the research backbone of Go Vegan. It is not a claim that every vegan diet, product, policy or argument is automatically ideal, sustainable or correct. Empirical claims are separated from ethical conclusions; peer-reviewed systematic reviews, meta-analyses and major studies are prioritized, while official primary sources are used where they are the appropriate authority. Each topic moves from a plain-English bottom line to the evidence, responsible wording, limitations, source and literal URL.

1. WHAT VEGANISM IS AND WHAT SCIENCE CAN ADDRESS

Veganism is broader than diet

BOTTOM LINE: Veganism is a philosophy and way of living that seeks to avoid animal exploitation and cruelty as far as possible and practicable. Food is one major part of veganism, but veganism also applies to clothing, testing, entertainment and other uses of animals.

WHAT THE EVIDENCE SHOWS: Science can answer factual questions that matter to veganism. For example: Can animals feel and experience the world? Do humans need animal foods to obtain essential nutrients? What effects does animal agriculture have on ecosystems and public health? Science can answer those questions. Whether avoidable animal exploitation is morally acceptable is an ethical question.

WHAT YOU CAN RESPONSIBLY SAY: Veganism is more than a diet. Science can establish many of the facts behind the vegan argument, while ethical reasoning addresses what we should do with those facts.

SOURCE: The Vegan Society: definition [Primary movement definition]
Definition of veganism.

URL: <https://www.vegansociety.com/go-vegan/definition-veganism>

2. HUMAN HEALTH & NUTRITIONAL ADEQUACY

Appropriately planned vegan diets can be nutritionally adequate across the life cycle

BOTTOM LINE: The Academy of Nutrition and Dietetics stated in its 2016 peer-reviewed position that appropriately planned vegetarian diets, including vegan diets, are healthful, nutritionally adequate, and appropriate during all stages of the life cycle.

WHAT THE EVIDENCE SHOWS: The position explicitly included pregnancy, lactation, infancy, childhood, adolescence, older adulthood and athletes. Adequacy still depends on planning: vitamin B12 requires a reliable fortified food or supplement, and other nutrients deserve the same deliberate attention they do in any diet.

WHAT YOU CAN RESPONSIBLY SAY: Appropriately planned vegan diets can be nutritionally adequate across the life cycle. Do not say every vegan diet is automatically adequate or that supplementation is never needed.

SOURCE: Melina et al. 2016 [Peer-reviewed Academy position]

Position of the Academy of Nutrition and Dietetics: Vegetarian Diets. J Acad Nutr Diet. DOI: 10.1016/j.jand.2016.09.025

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URL: <https://doi.org/10.1016/j.jand.2016.09.025>

SOURCE: Academy of Nutrition and Dietetics 2025 [Peer-reviewed current adult position]
Vegetarian Dietary Patterns for Adults: A Position of the Academy of Nutrition and Dietetics.

URL: <https://doi.org/10.1016/j.jand.2025.02.002>

Protein and amino acids

BOTTOM LINE: Plant foods contain all essential amino acids, and protein needs can be met with sufficient total intake and varied protein sources.

WHAT THE EVIDENCE SHOWS: Digestibility and amino-acid profiles differ among foods, so athletes, older adults or people eating low-calorie diets may benefit from deliberate protein planning. Routine 'protein combining' at every meal is unnecessary because the body maintains amino-acid pools across meals.

WHAT YOU CAN RESPONSIBLY SAY: Say 'plants provide all essential amino acids; adequacy depends on total intake and diet quality,' not 'all proteins are nutritionally identical.'

SOURCE: Mariotti & Gardner 2019 [Peer-reviewed]
Dietary Protein and Amino Acids in Vegetarian Diets, A Review.

URL: <https://pubmed.ncbi.nlm.nih.gov/31728571/>

Vitamin B12

BOTTOM LINE: B12 is the clearest non-negotiable nutrient in vegan nutrition: use a reliable supplement or appropriately fortified foods.

WHAT THE EVIDENCE SHOWS: B12 is synthesized by microorganisms. Modern dietary exposure is unreliable without supplementation/fortification, and deficiency can cause neurological and hematological harm.

WHAT YOU CAN RESPONSIBLY SAY: Never frame B12 supplementation as evidence that veganism 'fails.' Modern nutrition routinely uses fortification and supplements; the relevant question is whether requirements can be reliably met.

SOURCE: National Institutes of Health (NIH) Office of Dietary Supplements [Government evidence review]
Vitamin B12 Fact Sheet for Health Professionals.

URL: <https://ods.od.nih.gov/factsheets/VitaminB12-HealthProfessional/>

Iron, calcium, iodine, vitamin D, zinc and omega-3

BOTTOM LINE: These nutrients can be obtained on vegan diets, but several deserve deliberate planning because intake or bioavailability may be lower in some vegan populations.

WHAT THE EVIDENCE SHOWS: Iron from plants is absorbed differently from iron in meat, and absorption changes depending on the rest of the meal. Calcium depends on choosing calcium-rich or fortified foods. Iodine depends heavily on iodized salt, seaweed use and local food supply. Vitamin D depends on sunlight, fortified foods and supplements. Zinc absorption can be reduced by compounds called phytates. The omega-3 fat alpha-linolenic acid (ALA) comes from plant foods, while the longer-chain omega-3 fats eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) can be obtained directly from algae-based supplements.

WHAT YOU CAN RESPONSIBLY SAY: Do not answer nutrient questions with 'just eat plants.' Give the actual nutrient strategy.

SOURCE: Bakaloudi et al. 2021 [Peer-reviewed]
Intake and adequacy of the vegan diet: a systematic review of the evidence.

URL: <https://pubmed.ncbi.nlm.nih.gov/33341313/>

3. HEALTH OUTCOMES

Cardiometabolic outcomes in randomized trials

BOTTOM LINE: Vegan dietary interventions can improve several markers linked with heart and metabolic health, including low-density lipoprotein (LDL) cholesterol, body weight and measures of blood-sugar control.

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WHAT THE EVIDENCE SHOWS: Meta-analysis of randomized controlled trials reports beneficial changes in cardiometabolic markers, but interventions vary in food quality, energy intake and behavioral support. Benefits should not be attributed to the word 'vegan' independent of diet composition.

WHAT YOU CAN RESPONSIBLY SAY: Say well-designed vegan diets can improve cardiometabolic risk factors; do not claim veganism cures heart disease or guarantees disease prevention.

SOURCE: Termannsen et al. 2022 [Peer-reviewed]

Effects of vegan diets on heart and metabolic health: a systematic review and meta-analysis of randomized controlled trials.

URL: <https://pubmed.ncbi.nlm.nih.gov/35672940/>

Randomized study of identical twins

BOTTOM LINE: In an 8-week study of identical twins, the twin assigned to a healthy vegan diet had larger average reductions in low-density lipoprotein (LDL) cholesterol, fasting insulin and body weight than the twin assigned to a healthy omnivorous diet.

WHAT THE EVIDENCE SHOWS: Because identical twins share nearly all of their genes, comparing one twin with the other helps reduce the chance that genetic differences explain the result. However, the study lasted only eight weeks, included a relatively small number of people and compared two specifically designed healthy diets. It does not prove that every vegan diet is healthier than every omnivorous diet.

WHAT YOU CAN RESPONSIBLY SAY: This is useful experimental evidence that a healthy vegan diet can improve several short-term heart and metabolic risk markers. It does not tell us what happens over an entire lifetime.

SOURCE: Landry et al. 2023 [Peer-reviewed]

Cardiometabolic Effects of Omnivorous versus Vegan Diets in Identical Twins: A Randomized Clinical Trial.

URL: <https://doi.org/10.1001/jamanetworkopen.2023.44457>

Processed meat and cancer

BOTTOM LINE: The International Agency for Research on Cancer (IARC), the cancer agency of the World Health Organization, classifies processed meat as carcinogenic to humans and red meat as probably carcinogenic to humans.

WHAT THE EVIDENCE SHOWS: IARC categories describe how strong the evidence is that something can cause cancer. They do not tell us how large the risk is. Processed meat and tobacco are both in Group 1 because the evidence that they can cause cancer is strong, not because they are equally dangerous.

WHAT YOU CAN RESPONSIBLY SAY: State the classification accurately and avoid sensational comparisons.

SOURCE: Bouvard et al. 2015 [Peer-reviewed]

Carcinogenicity of consumption of red and processed meat. Lancet Oncology.

URL: <https://pubmed.ncbi.nlm.nih.gov/26514947/>

4. LIFE STAGES & ATHLETIC PERFORMANCE

Pregnancy

BOTTOM LINE: Current evidence does not justify saying vegan pregnancy is inherently unsafe, but the evidence base is small and careful nutrient management is important.

WHAT THE EVIDENCE SHOWS: A 2024 systematic review found only six eligible studies, lower intakes of several nutrients in vegan pregnancies, apparently adequate B12 status with supplementation, and lower birth weight in some comparisons. The authors emphasized insufficient power and heterogeneity.

WHAT YOU CAN RESPONSIBLY SAY: Say a vegan pregnancy requires deliberate nutrition and professional monitoring; do not claim the research is as mature as adult nutrition evidence.

SOURCE: Meulenbroeks et al. 2024 [Peer-reviewed]

The Association of a Vegan Diet during Pregnancy with Maternal and Child Outcomes: A Systematic Review.

URL: <https://pubmed.ncbi.nlm.nih.gov/39408296/>



Children and adolescents

BOTTOM LINE: Evidence in vegan children is growing but remains less certain than in adults; monitoring growth and key nutrients is warranted.

WHAT THE EVIDENCE SHOWS: Recent systematic reviews find mixed nutritional advantages and risks. B12 supplementation is essential; calcium, vitamin D, iron, zinc, iodine, omega-3 and adequate energy/protein deserve attention. Some meta-analyses report lower height or bone-mineral measures on average, while evidence quality and confounding remain important limitations.

WHAT YOU CAN RESPONSIBLY SAY: Do not use an adult position statement as if it settles pediatric nutrition. Say vegan childhood nutrition can be managed, but current evidence supports monitoring and careful planning.

SOURCE: Verduci et al. 2025 [Peer-reviewed]

Vegan diet and nutritional status in infants, children and adolescents: European Society for Paediatric Gastroenterology, Hepatology and Nutrition (ESPGHAN) position paper based on a systematic search.

URL: <https://pubmed.ncbi.nlm.nih.gov/40819279/>

SOURCE: Sutter et al. 2023 [Peer-reviewed]

Health aspects of vegan diets among children and adolescents: systematic review and meta-analyses.

URL: <https://pubmed.ncbi.nlm.nih.gov/37811643/>

Athletic performance

BOTTOM LINE: Current evidence does not show that well-planned plant-based diets impair athletic performance.

WHAT THE EVIDENCE SHOWS: A systematic review/meta-analysis found a moderate positive effect on aerobic performance, no significant effect on strength/power, and no overall performance disadvantage. Other reviews emphasize that energy, protein and micronutrient planning matter.

WHAT YOU CAN RESPONSIBLY SAY: Say vegan athletes can perform without an inherent disadvantage; do not promise that veganism automatically improves performance.

SOURCE: Damasceno et al. 2023 [Peer-reviewed]

Plant-based diets benefit aerobic performance and do not compromise strength/power performance: systematic review and meta-analysis.

URL: <https://pubmed.ncbi.nlm.nih.gov/37869973/>

SOURCE: Presti et al. 2024 [Peer-reviewed]

The Impact Plant-Based Diets Have on Athletic Performance and Body Composition: A Systematic Review.

URL: <https://pubmed.ncbi.nlm.nih.gov/38913935/>

5. ECOLOGY, CLIMATE, LAND & BIODIVERSITY

Overall environmental footprint

BOTTOM LINE: Diets containing fewer animal-source foods generally have lower greenhouse-gas emissions and land use, with vegan dietary patterns among the lowest-impact patterns studied.

WHAT THE EVIDENCE SHOWS: A 65-study scoping review found broad agreement for lower greenhouse-gas emissions, land use and biodiversity loss in plant-based patterns, while water and energy outcomes were more context-dependent. A 2025 systematic review of 120 studies also emphasized substantial large differences between countries, study methods and assumptions.

WHAT YOU CAN RESPONSIBLY SAY: Say the environmental advantage is strongest and most consistent for greenhouse-gas emissions and land; avoid claiming every plant food beats every animal food on every environmental metric.

SOURCE: Reinhardt et al. 2023 [Peer-reviewed]

The Environmental Sustainability of Plant-Based Dietary Patterns: A Scoping Review.

URL: <https://pubmed.ncbi.nlm.nih.gov/36809853/>

SOURCE: Environmental Impact of Current Diets 2025 [Peer-reviewed]

Environmental Impact of Current Diets and Alternative Dietary Scenarios Worldwide: A Systematic Review.

URL: <https://pubmed.ncbi.nlm.nih.gov/40459336/>

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Measured United Kingdom dietary impacts

BOTTOM LINE: In a large United Kingdom sample, environmental impacts increased with animal-food consumption; vegan diets had substantially lower impacts than high-meat diets across all measured indicators.

WHAT THE EVIDENCE SHOWS: The study linked dietary data from 55,504 people with data that estimate environmental impacts across a product's life cycle from more than 38,000 farms. Vegan diets were about 25% of high-meat diets for greenhouse-gas emissions and land use, 46% for water use, 27% for nutrient pollution of waterways (eutrophication) and 34% for biodiversity impact.

WHAT YOU CAN RESPONSIBLY SAY: Use the metric-specific numbers. Do not turn 'about 75% lower greenhouse-gas emissions and land use' into '75% lower everything.'

SOURCE: Scarborough et al. 2023 [Peer-reviewed]

Vegans, vegetarians, fish-eaters and meat-eaters in the United Kingdom show discrepant environmental impacts. Nature Food.

URL: <https://pubmed.ncbi.nlm.nih.gov/37474804/>

Climate mitigation and dietary shift

BOTTOM LINE: Major climate assessments identify dietary shifts toward more plant-based protein and away from high-emission animal foods as a meaningful food-system mitigation strategy.

WHAT THE EVIDENCE SHOWS: The IPCC concludes that diets high in plant protein and low in meat and dairy are associated with lower greenhouse-gas emissions. It also identifies co-benefits including land sparing and reduced nutrient losses, while emphasizing that appropriate diets and transition pathways vary by region and context.

WHAT YOU CAN RESPONSIBLY SAY: Plant-rich dietary shifts are an evidence-based climate mitigation tool, especially in high-consuming populations. Do not imply diet is the only climate solution or that identical dietary prescriptions fit every region.

SOURCE: Intergovernmental Panel on Climate Change (IPCC) AR6 WGIII, Chapter 12 [Major scientific assessment]

Food systems mitigation opportunities and dietary shift.

URL: <https://www.ipcc.ch/report/ar6/wg3/chapter/chapter-12/>

Global food production and land use

BOTTOM LINE: Animal products use a disproportionate share of global farmland relative to the calories and protein they provide, making land use one of the strongest environmental arguments for dietary change.

WHAT THE EVIDENCE SHOWS: Poore & Nemecek synthesized data from about 38,000 farms in 119 countries. Meat, aquaculture, eggs and dairy used about 83% of the world's farmland while providing about 18% of calories and 37% of protein. In their modeled no-animal-product scenario, global agricultural land use fell by about 76% while food production was maintained.

WHAT YOU CAN RESPONSIBLY SAY: Use the numbers with their scope: about 83% of farmland, 18% of calories, 37% of protein, and about 76% less agricultural land in the modeled no-animal-product scenario. This is a global scenario, not a claim that every hectare of grazing land can become cropland.

SOURCE: Poore & Nemecek 2018 [Peer-reviewed]

Reducing food's environmental impacts through producers and consumers. Science.

URL: <https://doi.org/10.1126/science.aag0216>

Biodiversity

BOTTOM LINE: Habitat conversion and agricultural land demand are major biodiversity pressures; livestock contributes through pasture and feed-crop land.

WHAT THE EVIDENCE SHOWS: Because animal agriculture uses extensive pasture plus feed-producing cropland, dietary shifts that reduce land demand can reduce habitat pressure. Biodiversity outcomes remain location-specific and cannot be inferred from carbon alone.

WHAT YOU CAN RESPONSIBLY SAY: Say reduced animal-food demand can reduce land-driven biodiversity pressure; do not claim diet is the only biodiversity driver.

SOURCE: Reinhardt et al. 2023 [Peer-reviewed]

Plant-based dietary patterns and environmental sustainability review.

URL: <https://pubmed.ncbi.nlm.nih.gov/36809853/>

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6. WATER, NUTRIENT POLLUTION & AIR QUALITY

Water use is context-dependent

BOTTOM LINE: Plant-rich diets generally reduce several environmental pressures, but water is more variable than land use or greenhouse gases.

WHAT THE EVIDENCE SHOWS: Water footprints depend strongly on crop, location, irrigation and whether green, blue or grey water is measured. Reviews find some sustainable diets can increase water use even while reducing greenhouse-gas emissions and land use.

WHAT YOU CAN RESPONSIBLY SAY: Do not claim vegan diets always use less water in every location. Use metric- and geography-specific evidence.

SOURCE: Aleksandrowicz et al. 2016 [Peer-reviewed]

The Impacts of Dietary Change on Greenhouse-Gas Emissions, Land, Water and Health: A Systematic Review.

URL: <https://pubmed.ncbi.nlm.nih.gov/27812156/>

SOURCE: Jarmul et al. 2020 [Peer-reviewed]

Climate change mitigation through dietary change: systematic review.

URL: <https://pubmed.ncbi.nlm.nih.gov/33897807/>

Manure and nutrient pollution

BOTTOM LINE: Concentrated animal production can create nitrogen, phosphorus, ammonia and waste-management burdens.

WHAT THE EVIDENCE SHOWS: Manure is a valuable fertilizer when appropriately managed, but nutrient surpluses and emissions can contribute to nutrient pollution of waterways (eutrophication), nitrate contamination and particulate-matter formation. Impacts depend on animal density, manure management and local nutrient balance.

WHAT YOU CAN RESPONSIBLY SAY: Do not describe manure as inherently 'toxic'; describe documented pollution pathways and management limits.

SOURCE: Gilbert et al. 2021 [Peer-reviewed]

Livestock intensification externalities including environmental degradation.

URL: <https://pubmed.ncbi.nlm.nih.gov/33573940/>

7. CROPS, FEED, SOY & THE 'CROP DEATHS' ARGUMENT

Feeding crops to animals is a conversion step

BOTTOM LINE: When human-edible crops are fed to livestock, much of their food energy is used for animal metabolism rather than returned as meat, milk or eggs.

WHAT THE EVIDENCE SHOWS: Global modeling has shown that redirecting suitable crops from feed and other uses toward direct human food can increase the calories available to people. Not all livestock feed is human-edible, and grazing systems differ, so the argument should focus on total land and resource use rather than pretending every kilogram of feed could be eaten by humans.

WHAT YOU CAN RESPONSIBLY SAY: Say animal agriculture often adds a trophic conversion step and substantial land demand; acknowledge non-human-edible feed and pasture.

SOURCE: Cassidy et al. 2013 [Peer-reviewed]

Redefining agricultural yields: from tonnes to people nourished per hectare.

URL: <https://doi.org/10.1088/1748-9326/8/3/034015>

Soy: what the 77% figure actually means

BOTTOM LINE: Roughly three-quarters of global soy is associated with animal feed, while only a small share is eaten directly as soy foods.

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WHAT THE EVIDENCE SHOWS: Soybeans are crushed into multiple co-products, especially high-protein soybean meal and soybean oil. Allocation statistics can therefore be described by mass, economic value or end use, and simplistic percentages can mislead. The widely cited ~77% feed figure largely reflects soybean meal used for livestock; direct foods such as tofu, soy milk, tempeh and edamame are a small share. Oil goes into cooking, processed foods, biodiesel and industrial uses.

WHAT YOU CAN RESPONSIBLY SAY: Say '~77% of global soy is used for animal feed, primarily as meal; direct human soy foods are a small share.' Do not say the remaining 16% is all 'industrial use.'

SOURCE: Ritchie & Roser / Our World in Data [Data synthesis]

Soy: allocation and drivers of deforestation, based on data from the United Nations Food and Agriculture Organization (FAO) and the Food Climate Research Network (FCRN).

URL: <https://ourworldindata.org/soy>

Crop deaths

BOTTOM LINE: Crop production can kill wild animals; veganism is not a claim that plant agriculture causes zero animal deaths.

WHAT THE EVIDENCE SHOWS: Reliable global counts of field-animal deaths are difficult because mortality varies by crop, region, farming practice and whether displacement/predation is counted. Animal agriculture does not eliminate crop impacts: many farmed animals consume crops, and livestock also occupies large pasture areas.

WHAT YOU CAN RESPONSIBLY SAY: The defensible argument is comparative: minimize avoidable harm and resource demand. Do not quote speculative crop-death numbers as settled science.

SOURCE: Fischer & Lamey 2018 [Peer-reviewed]

Field Deaths in Plant Agriculture. Journal of Agricultural and Environmental Ethics.

URL: <https://doi.org/10.1007/s10806-018-9733-8>

8. ANIMAL SENTIENCE, PAIN & COGNITION

Animal sentience

BOTTOM LINE: Sentience means having subjective experiences: in simple terms, there is something it feels like to be that individual. Sentient animals can have experiences that feel good or bad, including pain and pleasure.

WHAT THE EVIDENCE SHOWS: We cannot directly enter another being's mind, including another human's. Scientists therefore look for multiple kinds of evidence, including behavior, nervous-system activity, physiology, learning, memory and evolutionary similarities. When these independent lines of evidence point in the same direction, they provide evidence of sentience.

WHAT YOU CAN RESPONSIBLY SAY: Strong scientific evidence supports sentience in the vertebrate animals commonly used for food. An animal does not need human-level intelligence to be capable of feeling pain, fear, comfort or pleasure.

SOURCE: Browning & Birch 2022 [Peer-reviewed]

Animal sentience. Philosophy Compass. DOI: 10.1111/phc3.12822

URL: <https://pubmed.ncbi.nlm.nih.gov/35859762/>

SOURCE: Proctor et al. 2013 [Peer-reviewed]

Searching for Animal Sentience: A Systematic Review of the Scientific Literature. Animals.

URL: <https://pubmed.ncbi.nlm.nih.gov/26479539/>

Pain in farm animals

BOTTOM LINE: Farmed animals can experience pain, and scientists can measure evidence of that pain using behavior, body responses, injuries and validated pain-scoring methods.

WHAT THE EVIDENCE SHOWS: Research documents pain-related responses in mammals, birds and fish. Procedures such as castration and dehorning damage tissue and produce measurable behavioral and physiological changes associated with pain.

WHAT YOU CAN RESPONSIBLY SAY: We do not need to claim that another species experiences pain exactly as humans do. Multiple independent measurements show that farmed animals experience states consistent with pain and suffering.

SOURCE: Prunier et al. 2013 [Peer-reviewed]

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Identifying and monitoring pain in farm animals: a review. *Animal*.

URL: <https://pubmed.ncbi.nlm.nih.gov/23254122/>

SOURCE: Luna et al. 2023 [Peer-reviewed]

Measurement properties of pain scoring instruments in farm animals: a systematic review.

URL: <https://pubmed.ncbi.nlm.nih.gov/36662813/>

Fish sentience, emotion and pain

BOTTOM LINE: Fish are sentient vertebrate animals. Strong scientific evidence shows that fish can experience pain and fear, learn, remember, make decisions and enter positive or negative emotion-like states.

WHAT THE EVIDENCE SHOWS: Fish have brains, nervous systems and pain-detecting receptors called nociceptors. Experiments show learning, memory, decision-making, fear and anxiety-related behavior, responses to painful stimulation, and distinct emotion-like states that change with how fish perceive their environment. A 2022 review found hundreds of scientific studies relevant to fish sentience across 142 species and subspecies and concluded that the published literature contains abundant evidence for fish sentience.

WHAT YOU CAN RESPONSIBLY SAY: The scientific evidence supports treating fish as sentient animals capable of suffering. We should not claim that every human emotion has been demonstrated in fish; specific claims such as human-like guilt or romantic love require their own evidence.

SOURCE: Lambert et al. 2022 [Peer-reviewed review]

A Kettle of Fish: A Review of the Scientific Literature for Evidence of Fish Sentience. *Animals*.

URL: <https://pubmed.ncbi.nlm.nih.gov/35565608/>

SOURCE: Brown 2015 [Peer-reviewed review]

Fish intelligence, sentience and ethics. *Animal Cognition*.

URL: <https://pubmed.ncbi.nlm.nih.gov/24942105/>

SOURCE: Cerqueira et al. 2017 [Peer-reviewed experiment]

Cognitive appraisal of environmental stimuli induces emotion-like states in fish. *Scientific Reports*.

URL: <https://pubmed.ncbi.nlm.nih.gov/29030568/>

Plants and the 'plants feel pain' objection

BOTTOM LINE: Plants sense and respond to their environment, but a biological response to damage is not the same as consciously feeling pain.

WHAT THE EVIDENCE SHOWS: Plants use complex chemical and electrical signaling, but they do not have brains or neurons, and current evidence does not establish that plants consciously experience pain. Even if future evidence changed, feeding crops to farmed animals generally adds another resource-intensive step rather than avoiding plant agriculture.

WHAT YOU CAN RESPONSIBLY SAY: Plants respond to their environment. Current evidence does not show that they consciously feel pain. Avoid saying that plants are inert or do not respond to damage.

SOURCE: Taiz et al. 2020 [Peer-reviewed]

Plants Neither Possess nor Require Consciousness. *Trends in Plant Science*.

URL: <https://pubmed.ncbi.nlm.nih.gov/32880005/>

SOURCE: Draguhn et al. 2021 [Peer-reviewed]

Plants do not demonstrate animal-like consciousness.

URL: <https://pubmed.ncbi.nlm.nih.gov/34247030/>

9. FARMING, CONFINEMENT, TRANSPORT & SLAUGHTER

Animal agriculture creates recurring welfare risks

BOTTOM LINE: Animal agriculture exposes sentient animals to welfare risks from housing, handling, transport, painful procedures, disease, injury and slaughter.

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WHAT THE EVIDENCE SHOWS: Risk varies by species and production system; 'animal agriculture' is not one uniform practice. Welfare science nevertheless documents recurring hazards including pain, fear, heat stress, thirst, fatigue, lameness, injury and behavioral restriction.

WHAT YOU CAN RESPONSIBLY SAY: Argue from documented practices and welfare outcomes, not from the claim that every farm is identical or every worker is cruel.

SOURCE: Prunier et al. 2013 [Peer-reviewed]

Identifying and monitoring pain in farm animals.

URL: <https://pubmed.ncbi.nlm.nih.gov/23254122/>

SOURCE: European Food Safety Authority (EFSA) Animal Welfare [Scientific institution]

Scientific opinions and welfare assessments across farmed species.

URL: <https://www.efsa.europa.eu/en/topics/topic/animal-welfare>

Welfare improvement versus ending animal exploitation

BOTTOM LINE: Better welfare can reduce some suffering, but it is a different question from whether breeding, using and killing animals is justified when alternatives are available.

WHAT THE EVIDENCE SHOWS: Welfare research can compare housing, stunning, transport and pain mitigation. It cannot answer the ethical question of whether using animals as resources is acceptable in principle.

WHAT YOU CAN RESPONSIBLY SAY: Do not dismiss welfare reforms as meaningless; distinguish harm reduction from the vegan objection to exploitation itself.

SOURCE: Browning & Veit 2022 [Peer-reviewed]

The importance of end-of-life welfare.

URL: <https://doi.org/10.1007/s10806-022-09887-8>

10. PUBLIC HEALTH: ANTIMICROBIAL RESISTANCE, ZOOSES & FOOD SYSTEMS

Antimicrobial resistance

BOTTOM LINE: Antimicrobial use in food-animal systems can select for resistant bacteria and resistance genes that circulate among animals, people and environments.

WHAT THE EVIDENCE SHOWS: Antimicrobial resistance (antimicrobial resistance) happens when microbes such as bacteria evolve so that medicines designed to kill or control them no longer work effectively. It has many causes, including antimicrobial use in human medicine. Research also identifies food-animal production as an important setting where resistant organisms and resistance genes can develop and spread, especially when antimicrobial use and infection control are poorly managed.

WHAT YOU CAN RESPONSIBLY SAY: Antimicrobial use in livestock contributes to antimicrobial-resistance risk. Animal agriculture is not the only cause, and eliminating animal foods would not eliminate antimicrobial resistance by itself.

SOURCE: Gilbert et al. 2021 [Peer-reviewed]

Mitigating risks posed by livestock intensification: antimicrobial resistance and diseases that can pass between animals and humans.

URL: <https://pubmed.ncbi.nlm.nih.gov/33573940/>

SOURCE: Food-system antimicrobial-resistance umbrella review 2025 [Peer-reviewed]

Interconnections between the food system and antimicrobial resistance.

URL: <https://pubmed.ncbi.nlm.nih.gov/40746419/>

Diseases that can pass between animals and humans

BOTTOM LINE: Human-animal interfaces can create opportunities for pathogen emergence and transmission; livestock intensification can modify those risks.

WHAT THE EVIDENCE SHOWS: Zoonotic risk depends on species, density, biosecurity, land-use change, wildlife contact, trade and many other factors. Animal agriculture is one component of a broader disease-ecology system.

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WHAT YOU CAN RESPONSIBLY SAY: Say reducing risky animal-human interfaces can reduce some risks of diseases passing between animals and humans; do not claim veganism makes pandemics impossible.

SOURCE: Gilbert et al. 2021 [Peer-reviewed]

Review of livestock intensification, antimicrobial resistance and diseases that can pass between animals and humans.

URL: <https://pubmed.ncbi.nlm.nih.gov/33573940/>

11. FISHING & AQUACULTURE

Wild-capture fishing and bycatch

BOTTOM LINE: Fishing affects both target animals and non-target wildlife through capture, injury, mortality and ecosystem effects.

WHAT THE EVIDENCE SHOWS: Bycatch includes non-target fish, marine mammals, turtles, seabirds and invertebrates. Magnitudes vary dramatically by fishery and gear, and management can reduce bycatch without eliminating it.

WHAT YOU CAN RESPONSIBLY SAY: Use fishery-specific evidence when giving numbers; avoid one global bycatch percentage as if it applies everywhere.

SOURCE: Lewison et al. 2014 [Peer-reviewed]

Global patterns of marine mammal, seabird, and sea turtle bycatch reveal taxa-specific and cumulative megafauna hotspots.

URL: <https://doi.org/10.1073/pnas.1318960111>

Aquaculture does not automatically remove wild-fish dependence

BOTTOM LINE: Aquaculture can reduce pressure on some wild stocks, but feeds for many farmed species use fishmeal/fish oil alongside plant and other ingredients.

WHAT THE EVIDENCE SHOWS: Feed composition has changed substantially over time and differs by species. Carnivorous fish and shrimp have historically relied more heavily on marine ingredients than herbivorous/omnivorous species.

WHAT YOU CAN RESPONSIBLY SAY: Do not say all farmed fish are fed whole wild fish. Say aquaculture can retain links to capture fisheries through feed inputs.

SOURCE: Naylor et al. 2021 [Peer-reviewed]

A 20-year retrospective review of global aquaculture.

URL: <https://doi.org/10.1038/s41586-021-03308-6>

12. COMPANION ANIMAL NUTRITION

Dogs on plant-based diets

BOTTOM LINE: Available evidence indicates dogs can maintain health on appropriately formulated plant-based diets, but the evidence base is still much smaller than conventional pet-food research.

WHAT THE EVIDENCE SHOWS: A 12-month prospective study reported maintenance of clinical, hematological and nutritional health in dogs fed a complete commercial plant-based diet. Observational studies also exist, but owner-reported outcomes are vulnerable to bias.

WHAT YOU CAN RESPONSIBLY SAY: Say dogs require nutrients, not a particular ingredient category; complete formulation and veterinary monitoring matter.

SOURCE: Knight et al. 2024 [Peer-reviewed]

The impact of vegan diets on indicators of health in dogs: a 12-month prospective study.

URL: <https://pubmed.ncbi.nlm.nih.gov/38625934/>

Cats and obligate carnivory

BOTTOM LINE: 'Obligate carnivore' describes feline evolutionary physiology and nutrient requirements; it does not by itself prove every required nutrient must come from slaughtered animal tissue.

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WHAT THE EVIDENCE SHOWS: Cats require nutrients such as taurine, preformed vitamin A and arachidonic acid in appropriate amounts/forms. A large observational study found no statistically significant overall health disadvantage in vegan-fed cats, but observational evidence cannot establish equivalence and long-term controlled data remain limited.

WHAT YOU CAN RESPONSIBLY SAY: Say appropriately formulated diets are the scientific question. Do not claim current evidence proves vegan diets are superior or risk-free for cats.

SOURCE: Knight et al. 2023 [Peer-reviewed]

Vegan versus meat-based cat food: Guardian-reported health outcomes in 1,369 cats.

URL: <https://pubmed.ncbi.nlm.nih.gov/37703240/>

SOURCE: Domínguez-Oliva et al. 2023 [Peer-reviewed]

The Impact of Vegan Diets on Indicators of Health in Dogs and Cats: A Systematic Review.

URL: <https://pubmed.ncbi.nlm.nih.gov/36669053/>

13. COST, FOOD ACCESS & ECONOMICS

Can plant-based diets cost less?

BOTTOM LINE: Staple-centered plant-based diets can be inexpensive, but vegan affordability depends on geography, food prices, cooking facilities, time and access.

WHAT THE EVIDENCE SHOWS: Computer-modeling studies and real dietary intervention studies in high-income settings often find plant-forward or vegan patterns can reduce food costs, especially when based on grains, legumes and other staples rather than specialty substitutes. This cannot be generalized to every household or region.

WHAT YOU CAN RESPONSIBLY SAY: Say vegan eating can be affordable and often cheaper in high-income settings; do not dismiss genuine access constraints.

SOURCE: Springmann et al. 2021 [Peer-reviewed]

The global and regional costs of healthy and sustainable dietary patterns.

URL: [https://doi.org/10.1016/S2542-5196\(21\)00251-5](https://doi.org/10.1016/S2542-5196(21)00251-5)

SOURCE: Kahleova et al. 2023 [Peer-reviewed]

Food costs in a randomized low-fat vegan dietary intervention.

URL: <https://pubmed.ncbi.nlm.nih.gov/37669055/>

Costs that are not included in the price of food

BOTTOM LINE: Food prices do not necessarily include climate, health, pollution or biodiversity costs.

WHAT THE EVIDENCE SHOWS: Global modeling studies estimate large potential health and climate economic benefits from dietary shifts, but these are estimates from computer models that depend on assumptions about future diets, disease, prices and policy.

WHAT YOU CAN RESPONSIBLY SAY: Use economic scenarios as models, not observed savings or guaranteed outcomes.

SOURCE: Springmann et al. 2016 [Peer-reviewed]

Analysis and valuation of the health and climate change cobenefits of dietary change.

URL: <https://doi.org/10.1073/pnas.1523119113>

14. WORKERS, COMMUNITIES & SOCIAL DIMENSIONS

Slaughterhouse and meat-processing work

BOTTOM LINE: Animal-product supply chains also affect human workers through injury, repetitive strain, hazardous machinery, biological exposure and psychosocial stress.

WHAT THE EVIDENCE SHOWS: Worker outcomes vary by country, plant, job and regulation. Research on mental-health effects is less uniform than research on physical occupational hazards, so claims about trauma should be carefully sourced rather than assumed.

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WHAT YOU CAN RESPONSIBLY SAY: Include workers in the ethical picture without using them as rhetorical props. Separate well-established occupational hazards from emerging psychosocial research.

SOURCE: Leibler et al. 2017 [Peer-reviewed]

Prevalence of serious psychological distress among slaughterhouse workers at a United States beef packing plant.

URL: <https://doi.org/10.3233/WOR-172543>

15. FARMERS, RURAL LIVELIHOODS & TRANSITION

A vegan transition is a production transition, not a claim that farmers should be abandoned

BOTTOM LINE: Reducing animal agriculture would shift demand, land use, infrastructure and employment; transition policy matters.

WHAT THE EVIDENCE SHOWS: Science can model land, environmental and economic changes, but specific transition programs are policy and implementation questions. Existing projects demonstrate that some farmers can move from animal production toward crops, horticulture, renewable energy, conservation or other revenue streams; they do not prove every farm can make the same transition.

WHAT YOU CAN RESPONSIBLY SAY: Pair environmental arguments with a just-transition framework: technical support, finance, market development, debt relief where appropriate, retraining and rural investment.

SOURCE: Transfarming [Practical transition resource]

Farmer transition program and case studies.

URL: <https://transfarmingproject.org/>

SOURCE: Rowdy Girl Sanctuary [Practical transition resource]

Rancher Advocacy Program.

URL: <https://rowdygirlsanctuary.org/>

16. COMMON ARGUMENTS: WHAT SCIENCE DOES: AND DOES NOT: SETTLE

“Animals eat animals.”

BOTTOM LINE: Predation in nature is a biological fact, not a scientific demonstration of what humans ought to do.

WHAT THE EVIDENCE SHOWS: Evolution explains behaviors; it does not create moral obligations. Humans routinely decline to imitate many natural behaviors.

WHAT YOU CAN RESPONSIBLY SAY: Treat this as an ethical/logic argument, not a nutrition argument.

“Humans are omnivores.”

BOTTOM LINE: Humans can digest both plant and animal foods; capability does not establish necessity.

WHAT THE EVIDENCE SHOWS: The nutritional question is whether requirements can be met without animal foods. Adult nutrition evidence shows they can be with appropriate planning.

WHAT YOU CAN RESPONSIBLY SAY: Separate 'can eat' from 'must eat.'

“It is natural / traditional.”

BOTTOM LINE: Naturalness and tradition describe origin or history, not whether an action is necessary or ethically justified.

WHAT THE EVIDENCE SHOWS: Science can document historical diets and biological adaptation, but the moral inference requires an ethical premise.

WHAT YOU CAN RESPONSIBLY SAY: Do not deny history; ask what follows ethically from it today.

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“Veganism is not perfect.”

BOTTOM LINE: No food system is harm-free. Veganism is a harm/exploitation-reduction framework, not a claim of zero impact.

WHAT THE EVIDENCE SHOWS: Agriculture has unavoidable trade-offs. Comparative evidence on land, emissions and animal use is more informative than demanding perfection from only one option.

WHAT YOU CAN RESPONSIBLY SAY: Compare realistic alternatives rather than veganism with an impossible zero-harm standard.

17. CLAIMS THIS LIBRARY DOES NOT SUPPORT

“A vegan diet is automatically healthy.”: False. A diet can be vegan and nutritionally poor.

“Veganism causes zero animal deaths.”: False. Crop production, transport, construction and other human activities can harm animals.

“Plants do not respond to injury.”: False. Plants have complex signaling; the unsupported leap is equating that with conscious pain.

“All soy not eaten directly by humans is industrial waste.”: False. Soy processing produces meal and oil with multiple linked markets.

“Vegan diets use 75% less water.”: Too broad. Water results are much more context-dependent than greenhouse-gas emissions and land use.

“Animal agriculture is the sole cause of antimicrobial resistance or diseases that can pass between animals and humans.”: False.

“Every child, pregnant person, athlete, dog or cat can switch diets without planning.”: Unsupported and potentially unsafe.

“One study proves veganism.”: Science works through converging evidence, replication, study quality and uncertainty.

“Ethics is scientifically proven.”: Science establishes empirical premises; moral conclusions require ethical reasoning.

18. RESEARCH ROADMAP & EVIDENCE GAPS

Long-term controlled studies of vegan diets across pregnancy, childhood, older age and diverse populations.

Better nutrient-status data separating well-planned vegan diets from self-selected restrictive diets.

More direct comparisons of whole-food vegan patterns versus highly processed vegan patterns.

More geographically diverse environmental assessments, especially water, biodiversity and low-/middle-income settings.

Better field-animal mortality measurement across crop systems and farming practices.

Long-term controlled companion-animal nutrition studies, particularly cats.

Worker and community-health research that distinguishes physical hazards, psychosocial outcomes and socioeconomic context.

Transition research on farmer income, debt, land suitability, rural employment and policy design.

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19. CORE REFERENCE INDEX

The references below are intentionally repeated from the claim-specific source blocks so the library can also be searched quickly by author/topic.

Nutrition adults: <https://doi.org/10.1016/j.jand.2025.02.002>
Vegan randomized controlled trial meta-analysis: <https://pubmed.ncbi.nlm.nih.gov/35672940/>
Twin randomized controlled trial: <https://doi.org/10.1001/jamanetworkopen.2023.44457>
Protein review: <https://pubmed.ncbi.nlm.nih.gov/31728571/>
Pregnancy systematic review: <https://pubmed.ncbi.nlm.nih.gov/39408296/>
Children ESPGHAN 2025: <https://pubmed.ncbi.nlm.nih.gov/40819279/>
Athletic performance meta-analysis: <https://pubmed.ncbi.nlm.nih.gov/37869973/>
Environmental scoping review: <https://pubmed.ncbi.nlm.nih.gov/36809853/>
United Kingdom environmental impacts: <https://pubmed.ncbi.nlm.nih.gov/37474804/>
Poore & Nemecek: <https://doi.org/10.1126/science.aag0216>
Dietary environmental systematic review: <https://pubmed.ncbi.nlm.nih.gov/27812156/>
Animal sentience review: <https://pubmed.ncbi.nlm.nih.gov/35859762/>
Farm-animal pain review: <https://pubmed.ncbi.nlm.nih.gov/23254122/>
Fish pain/stress: <https://pubmed.ncbi.nlm.nih.gov/25000797/>
Plant consciousness critique: <https://pubmed.ncbi.nlm.nih.gov/32880005/>
Livestock antimicrobial resistance and diseases that can pass between animals and humans: <https://pubmed.ncbi.nlm.nih.gov/33573940/>
Dog prospective study: <https://pubmed.ncbi.nlm.nih.gov/38625934/>
Cat observational study: <https://pubmed.ncbi.nlm.nih.gov/37703240/>
Dog/cat systematic review: <https://pubmed.ncbi.nlm.nih.gov/36669053/>
Cost modeling: [https://doi.org/10.1016/S2542-5196\(21\)00251-5](https://doi.org/10.1016/S2542-5196(21)00251-5)

